Introduction to Artificial Intelligence

Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks commonly associated with human intelligence. These tasks include understanding language, recognizing patterns, solving problems, and making predictions.

Machine Learning

Machine Learning (ML) is a subset of AI. It enables computer systems to learn patterns from data and improve their performance on a task without being explicitly programmed with every rule. Common learning settings include supervised learning, unsupervised learning, and reinforcement learning.

Deep Learning

Deep Learning is a subset of machine learning that uses neural networks with multiple layers. It is used in applications such as image recognition, speech processing, language understanding, and recommendation systems.

Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to process and work with human language. Examples include text classification, translation, summarization, question answering, and conversational assistants.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. A retrieval component finds relevant passages from a knowledge source, and the language model uses those passages as context when generating a response. RAG can help ground answers in provided documents, although retrieval quality and model behavior still need evaluation.